

Interpreting the FAIR principles in the age of infrastructure and automation

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Question

- the FAIR principles were formulated in 2016 with the aim of improving the re-use of scientific data



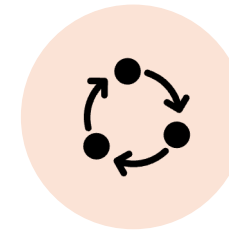
Findable



Accessible



Interoperable



Reusable

- the world has changed in the past ten years
- is there more to be said about their interpretation?

The changes

- data re-use is happening – including the use of real-world data in research
 - some of the data and metadata doesn't 'belong' to the researcher
- research is making greater use of shared infrastructure
 - including mandatory use of data spaces and secure processing environments
- AI technologies
 - automatic interpretation of natural language is now routine

To be findable

- F1** (meta)data are assigned a globally unique and persistent identifier
- F2** data are described with rich metadata (defined by R1 below)
- F3** metadata clearly and explicitly include the identifier of the data it describes
- F4** (meta)data are registered or indexed in a searchable resource

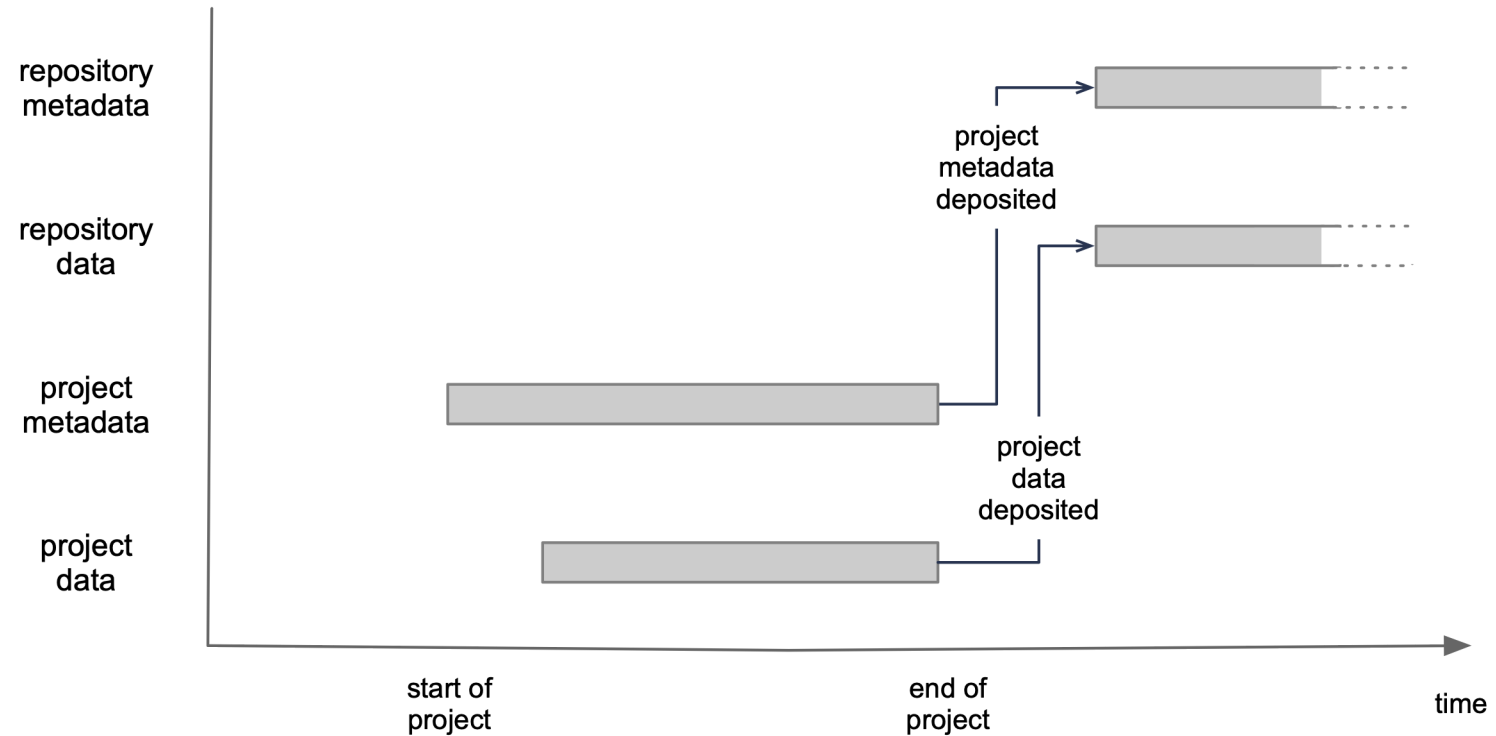
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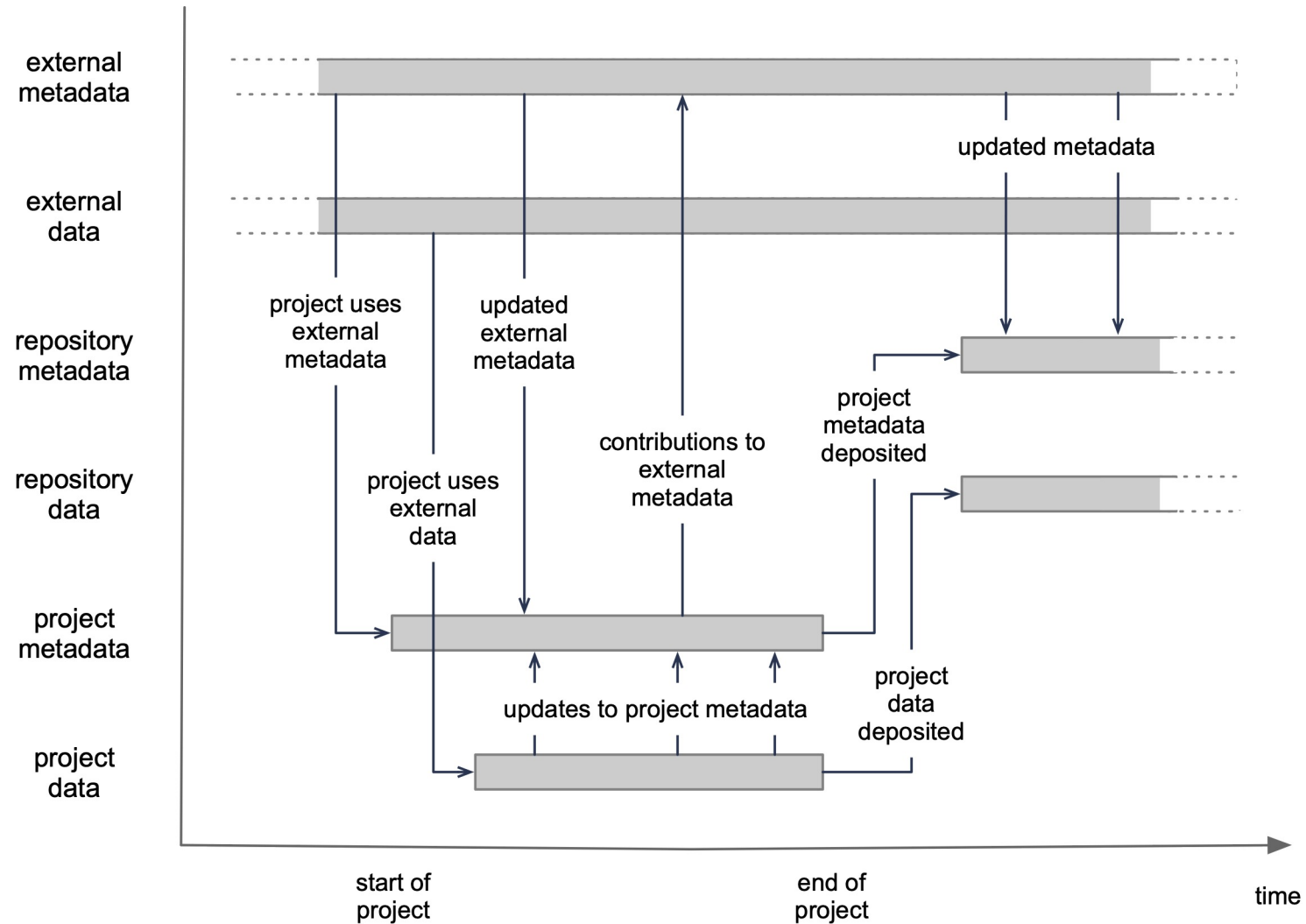


where does this come from, and who is responsible for it?

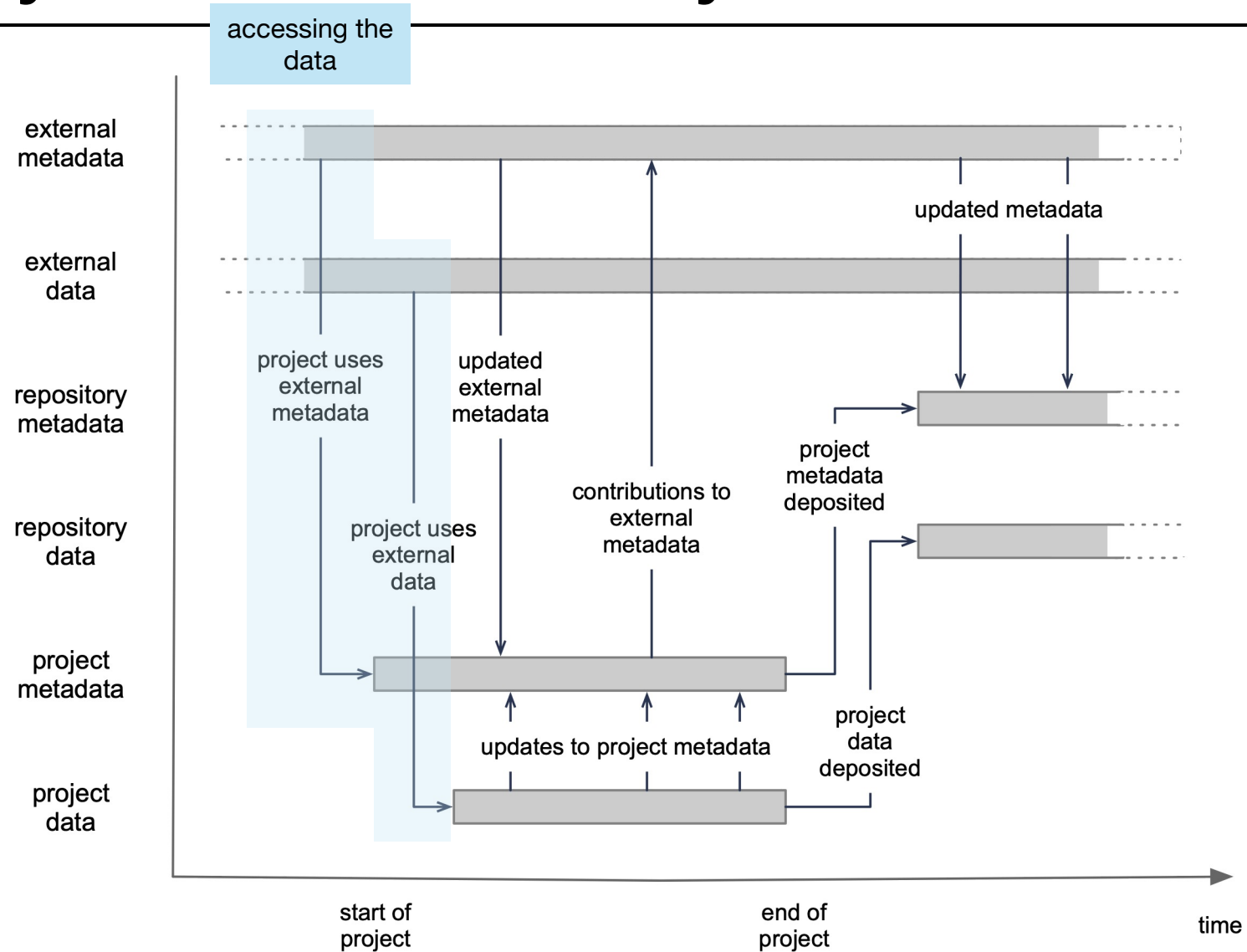
Primary metadata



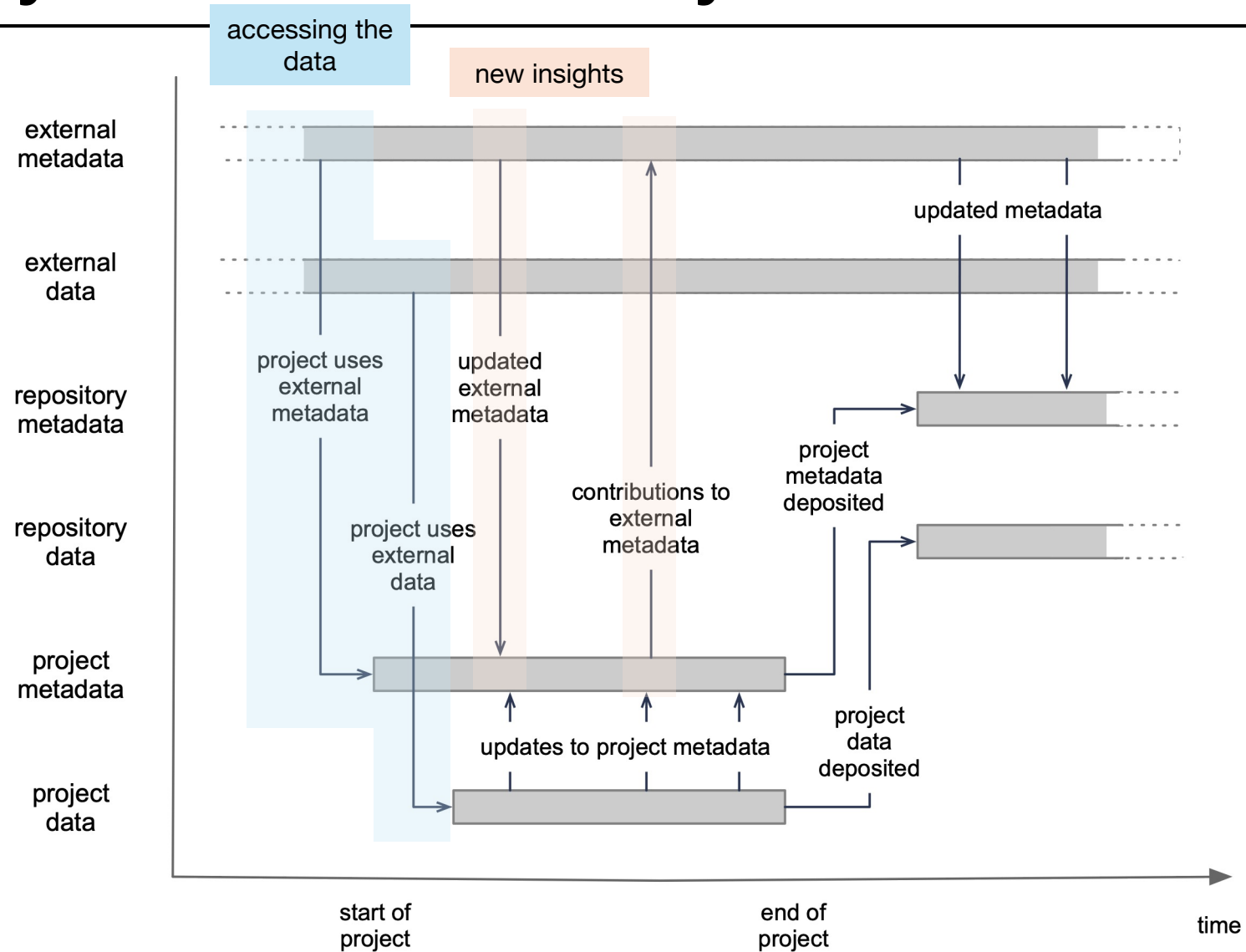
Primary and secondary metadata



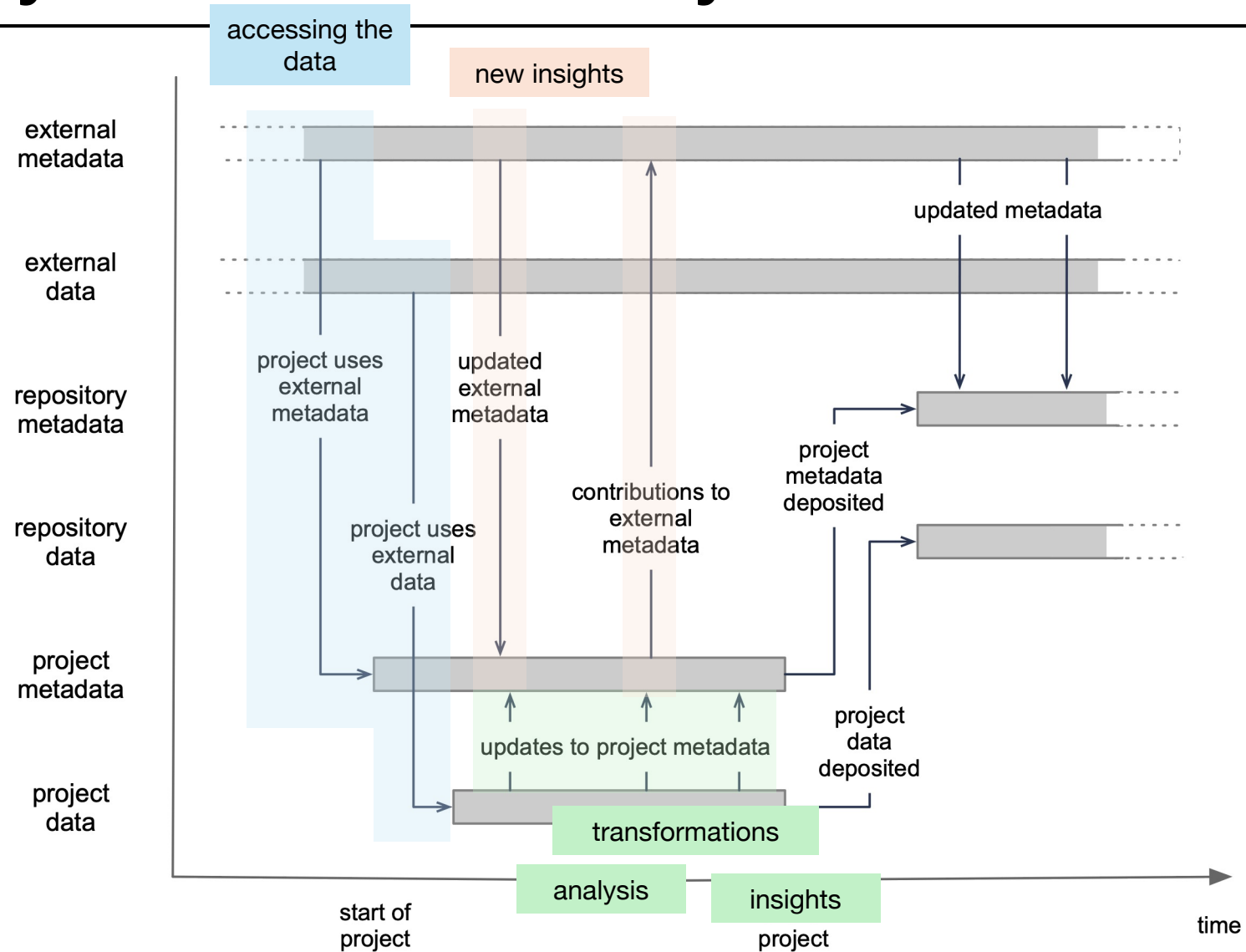
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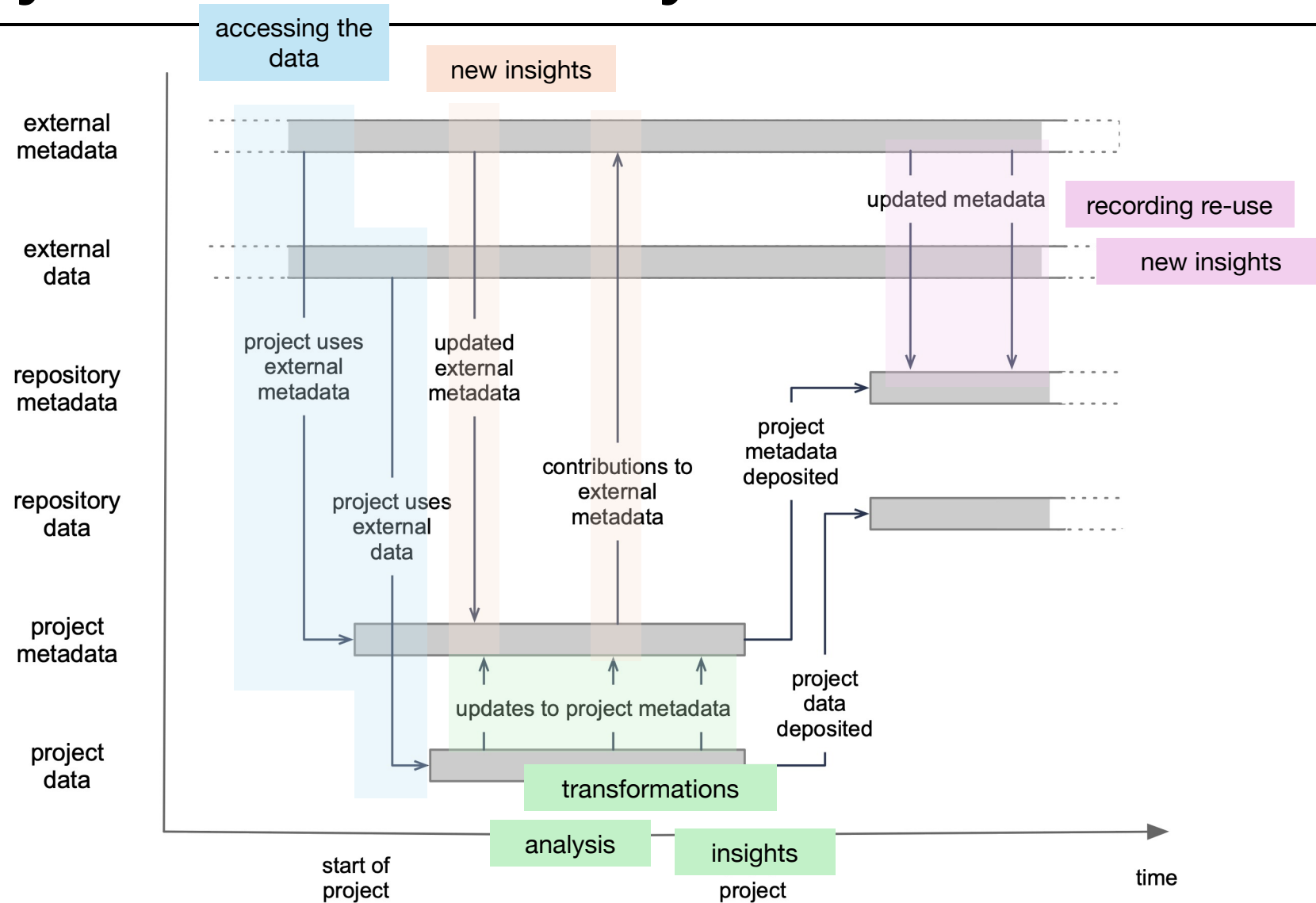
Primary and secondary metadata



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To be accessible

- A1** (meta)data are retrievable by their identifier using a standardized communications protocol
 - A1.1** the protocol is open, free, and universally implementable
 - A1.2** the protocol allows for an authentication and authorization procedure, where necessary
- A2** metadata are accessible, even when the data are no longer available

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A1.1 the protocol is open, free, and universally implementable

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A2 metadata are accessible, even when the data are no longer available

you may need permission from different organisations to access different parts of the data and metadata.



To be interoperable

- I1** (meta)data use a formal, accessible, shared, and broadly applicable language for knowledge representation
- I2** (meta)data use vocabularies that follow FAIR principles
- I3** (meta)data include qualified references to other (meta)data

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how far, and how fast, does this go?

this is now more about the semantics
than the syntax

To be re-usable

R1 (meta)data are richly described with a plurality of accurate and relevant attributes

R1.1 (meta)data are released with a clear and accessible data usage license

R1.2 (meta)data are associated with detailed provenance

R1.3 (meta)data meet domain-relevant community standards

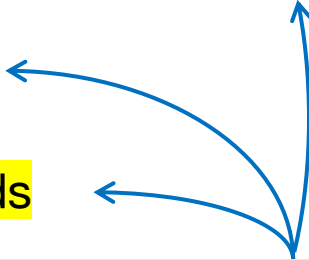
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the distinction between semantic interoperability
and re-usability is increasingly blurred

Summary

- some of the data and metadata may belong to a different organisation
- the relevant metadata may be extended over time, beyond the lifetime of the project
- interoperability and re-usability are increasingly indistinguishable
- the role of the infrastructure is increasing
 - not simply recording information that might inform re-use, but recording the fact of re-use, and learning from it

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